

# Advanced Python Programming For Data Science

Chapter 4: Applied Statistics and Exploratory Analysis

# Chapter Overview

This chapter focuses on:

- Understanding data using statistical measures
- Analyzing relationships and patterns in data
- Applying probability and hypothesis testing
- Exploring data systematically before modeling

The emphasis is on **interpreting data**, not just computing values.

Duration: 7 hours

Marks Weightage: 10 Marks

# Learning Outcomes

After completing this chapter, students will be able to:

- Compute and interpret key statistical measures
- Apply probability concepts and hypothesis testing
- Analyze trends and relationships using regression
- Perform exploratory data analysis (EDA)
- Automate EDA workflows using Python

# Chapter Structure



4.1 Statistical measures: Correlation, covariance, skewness, kurtosis



4.2 Probability review, sampling, and hypothesis testing



4.3 Regression and trend analysis using stats models



4.4 Exploratory data analysis (EDA) using descriptive and inferential methods



4.5 Automation of EDA workflows using Python

# Why Chapter 4 Is Important

- Raw data alone does not provide insights
- Statistics help **summarize, compare, and explain data**
- EDA helps detect:
  - Patterns
  - Outliers
  - Relationships
- Statistical understanding is essential before:
  - Machine learning
  - Visualization
  - Decision-making

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# Relationship to Previous Chapters

Chapter 4 builds directly on concepts learned in earlier chapters:

- **Chapter 1 – Advanced Python Concepts**  
Python fundamentals, functions, and OOP provide the programming foundation required to implement statistical analysis.
- **Chapter 2 – Data Sources and APIs**  
Data collected from files, databases, APIs, and web sources serves as the input for statistical analysis and EDA.
- **Chapter 3 – Data Wrangling and Transformation**  
Cleaned, transformed, and structured data is essential before applying statistics and exploratory analysis.
- **Chapter 4 – Applied Statistics and EDA**  
Uses prepared data to:
  - Summarize distributions
  - Identify relationships and patterns
  - Draw meaningful insights

**Concept Flow:**

**Python Basics → Data Collection → Data Preparation → Statistical Analysis & EDA**

## 4.1 Statistical measures: Correlation, covariance, skewness, kurtosis

### What Are Statistical Measures?

**Statistical measures** are numerical values used to summarize, describe, and understand data.

They help us:

- Understand relationships between variables
- Measure variability and distribution shape
- Interpret data beyond raw values

Statistical measures are a foundation of **exploratory data analysis (EDA)**.



# Why Statistical Measures Are Important

Statistical measures help to:

- Identify patterns and trends
- Detect relationships between variables
- Understand data distribution and spread
- Support data-driven decision making

Without statistical measures, data analysis remains descriptive and incomplete.



# Correlation

**Correlation** measures the **strength and direction of the relationship** between two variables.

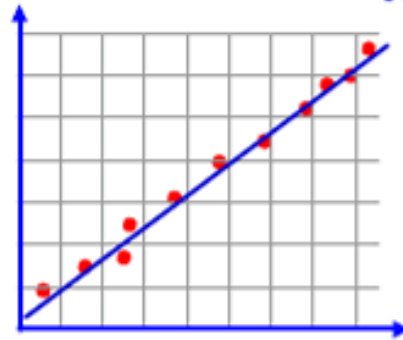
Key points:

- Values range from **-1 to +1**
- Positive correlation → variables move together
- Negative correlation → variables move in opposite directions
- Zero correlation → no linear relationship

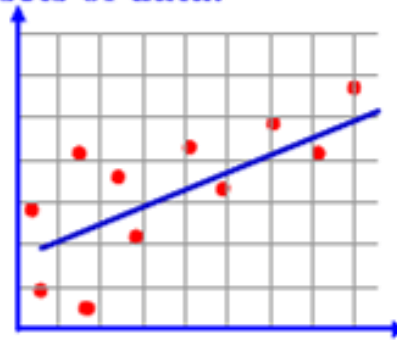
Correlation indicates **association**, not causation.

## SCATTERPLOTS & CORRELATION

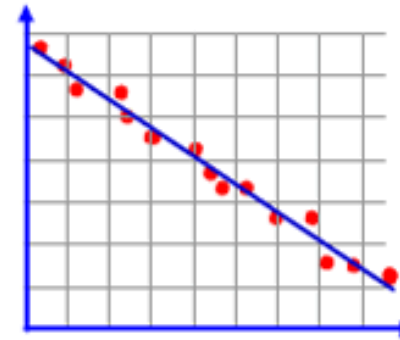
Correlation - indicates a relationship (connection) between two sets of data.



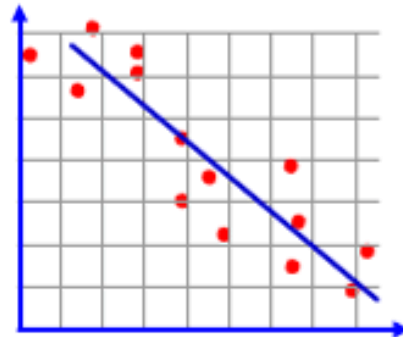
**Strong positive correlation**



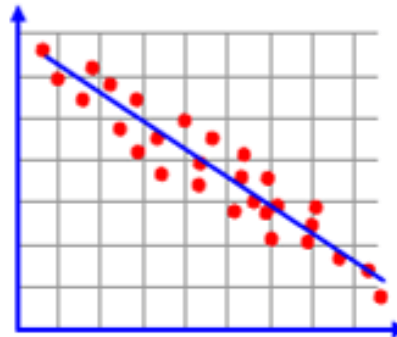
**Weak positive correlation**



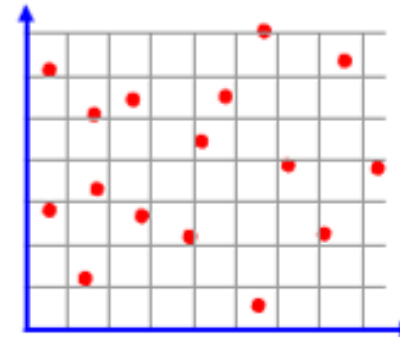
**Strong negative correlation**



**Weak negative correlation**



**Moderate negative correlation**



**No correlation**

# Covariance

**Covariance** measures how **two variables change together**.

Key points:

- Positive covariance → variables increase or decrease together
- Negative covariance → one increases while the other decreases
- Magnitude is not standardized, making interpretation difficult

Covariance is mainly used as a **foundation for correlation**.

# Correlation vs Covariance

| Aspect              | Correlation             | Covariance                |
|---------------------|-------------------------|---------------------------|
| Scale               | Standardized (−1 to +1) | Not standardized          |
| Interpretation      | Easy                    | Difficult                 |
| Use                 | Relationship strength   | Direction of relationship |
| Dependency on units | No                      | Yes                       |

# Skewness

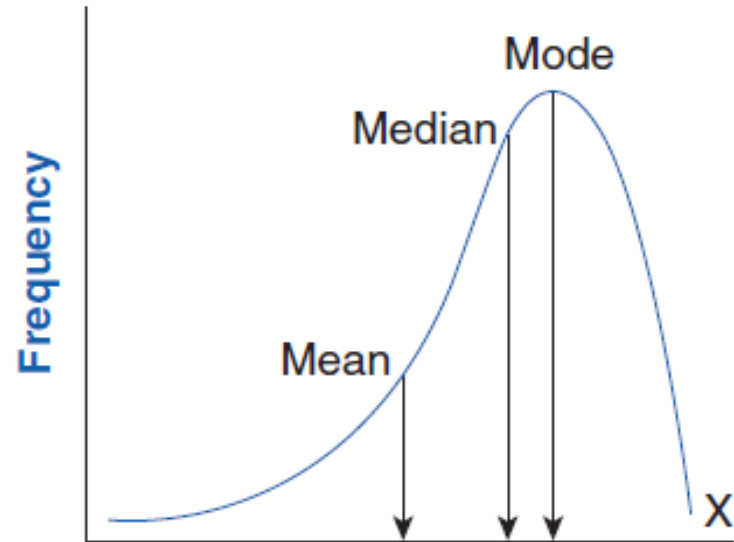
**Skewness** measures the **asymmetry of a data distribution**.

Types:

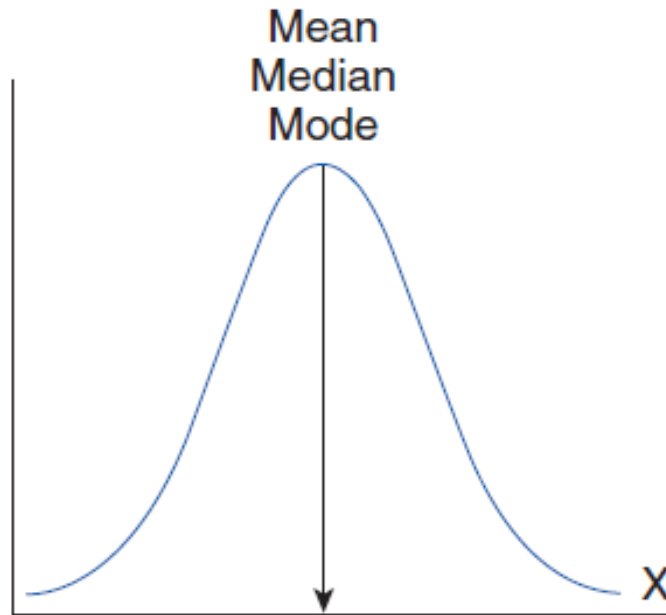
- Positive skew → long tail on the right
- Negative skew → long tail on the left
- Zero skew → symmetric distribution

Skewness helps identify **bias in data distribution**.

(a) Negatively skewed

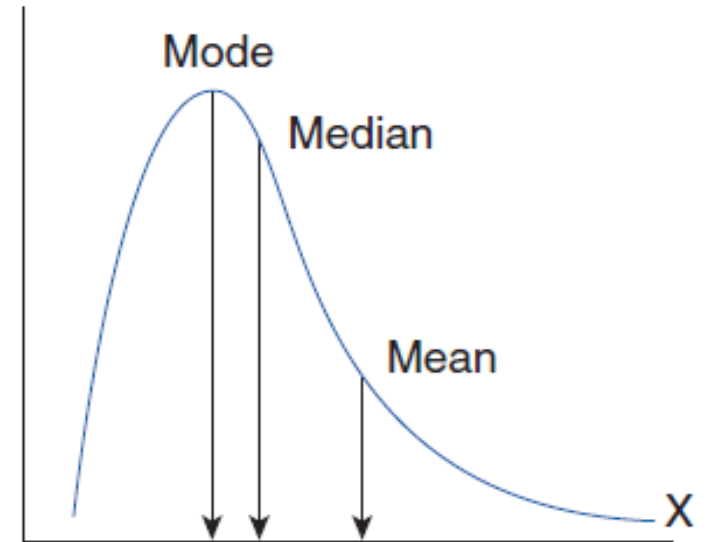


(b) Normal (no skew)



The normal curve represents a perfectly symmetrical distribution

(c) Positively skewed



# Kurtosis

**Kurtosis** measures the **tailedness or peakedness** of a distribution.

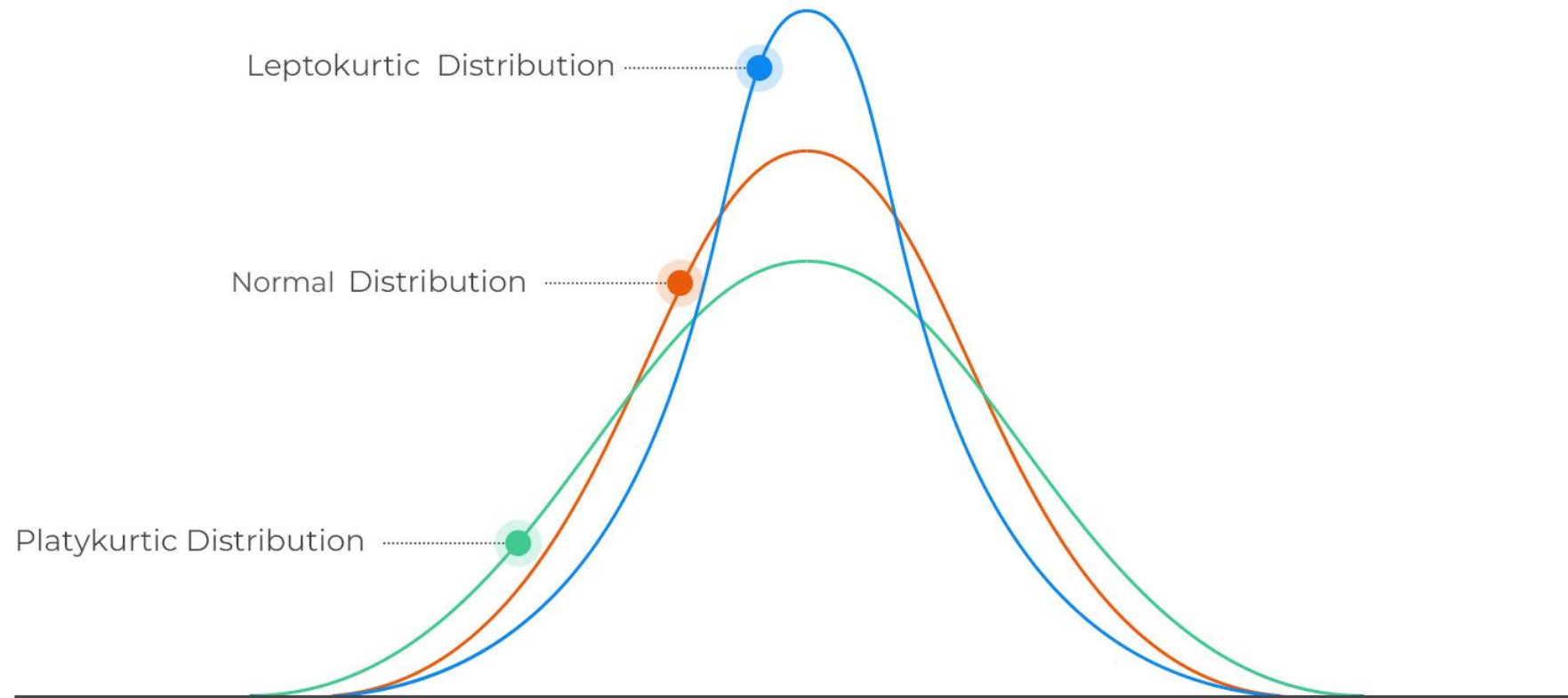
Key ideas:

- High kurtosis → heavy tails, more outliers
- Low kurtosis → light tails, fewer outliers

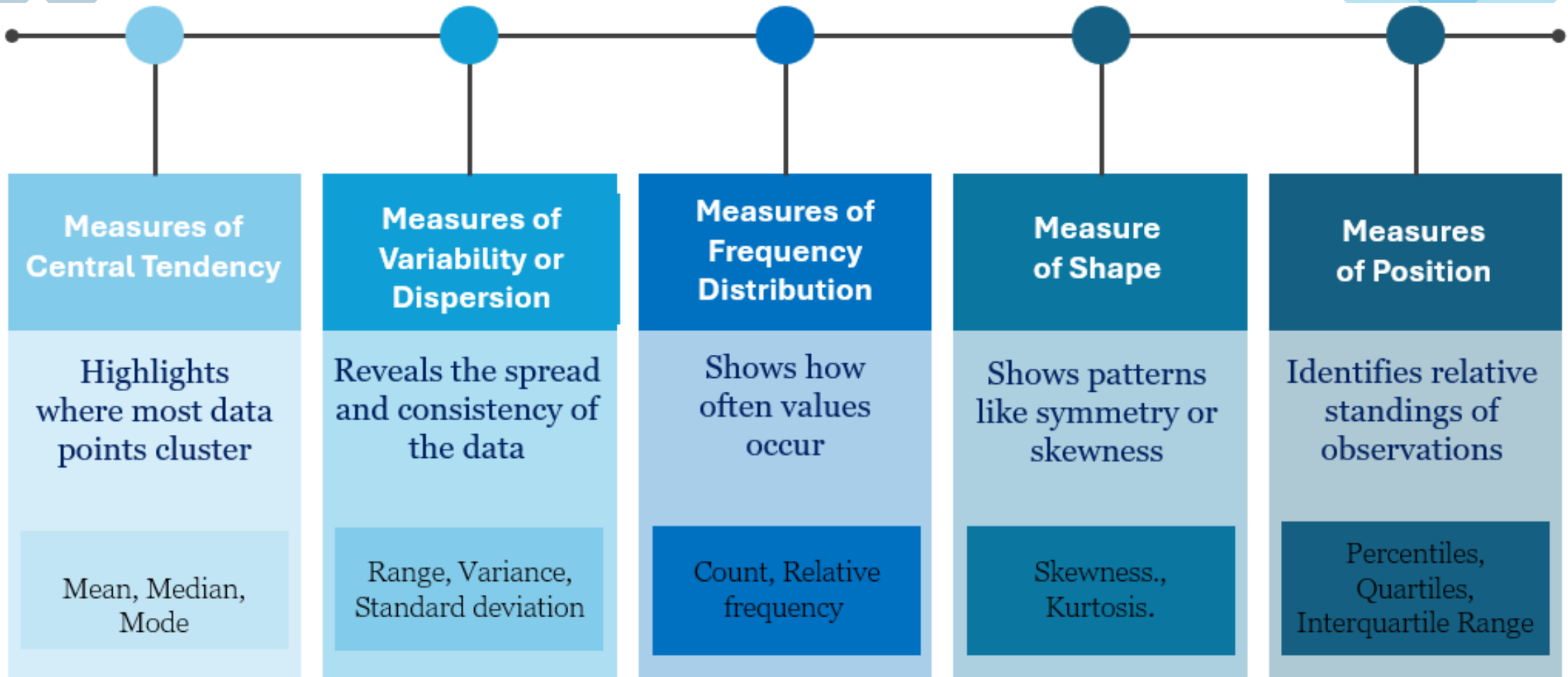
Kurtosis helps assess **outlier presence and distribution shape**.



## Kurtosis







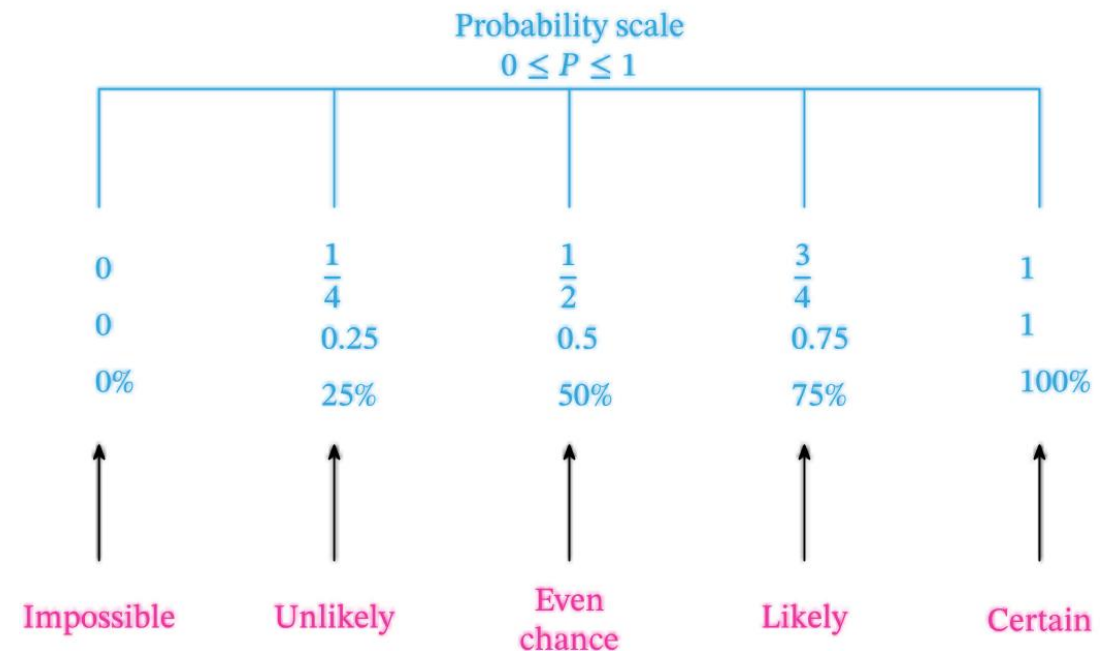
## 4.2 Probability Review, Sampling, and Hypothesis Testing

**Probability** is a measure of the **likelihood of an event occurring**.

Key points:

- Probability values range from **0 to 1**
- 0 → impossible event
- 1 → certain event
- Used to model **uncertainty** in real-world data

Probability forms the **foundation of statistical inference**.



# Probability Distribution

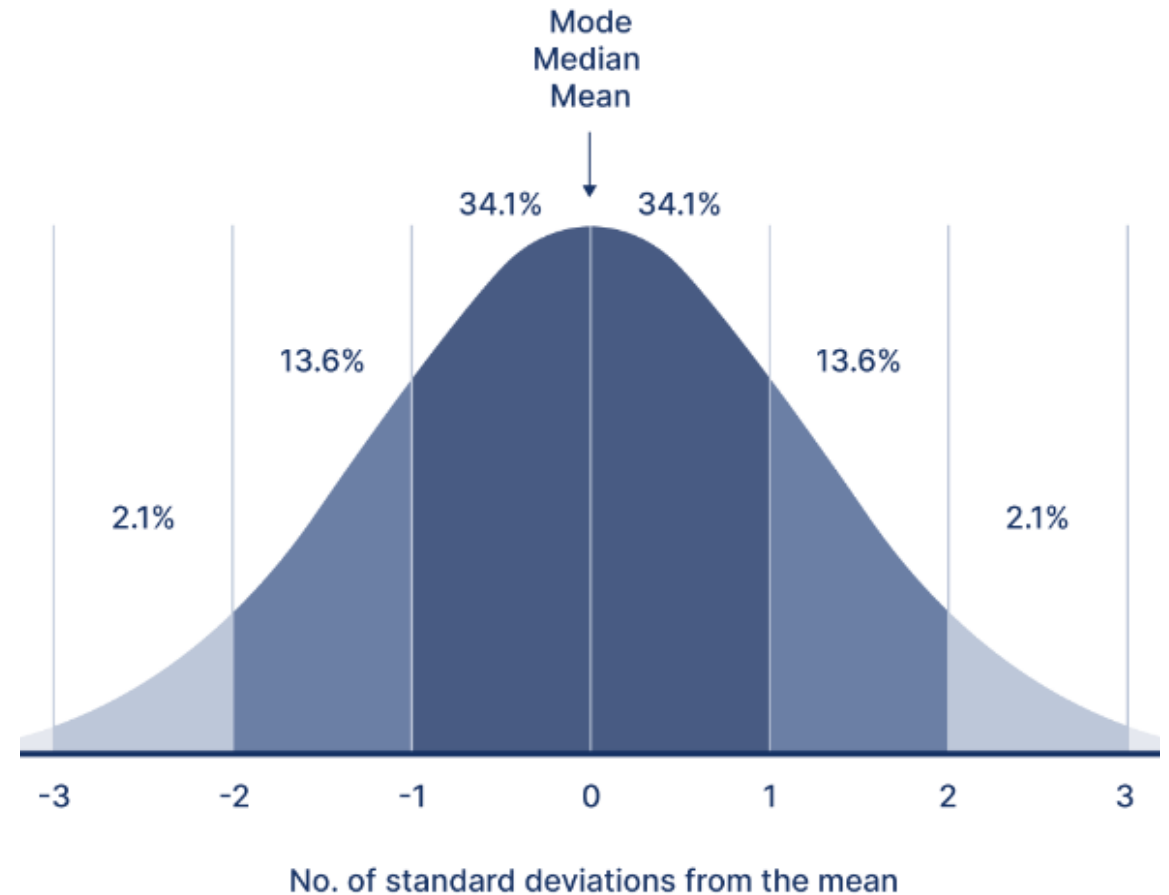
A **probability distribution** describes how values of a variable are **distributed over possible outcomes**.

Key points:

- Shows frequency or likelihood of values
- Helps understand data spread and central tendency
- Common distributions include normal distribution

Distributions are essential for **statistical modeling and hypothesis testing**.

## Standard normal distribution



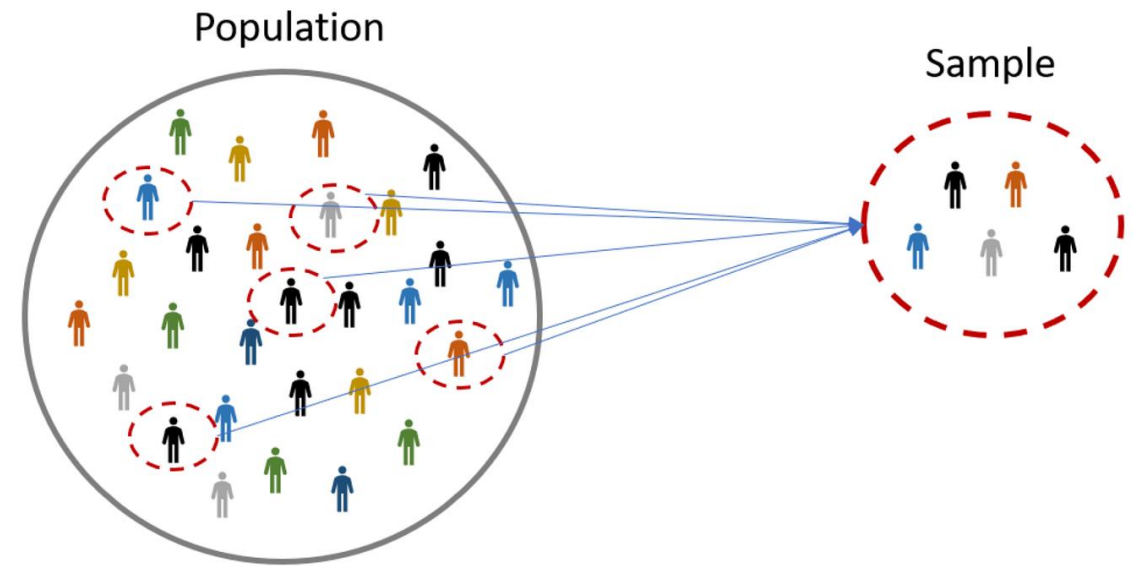
# Sampling

**Sampling** is the process of selecting a **subset (sample)** from a **population** to draw conclusions about the entire population.

Key points:

- Studying the full population is often impractical
- Samples reduce cost and computation
- Conclusions depend on sample quality

Sampling enables **generalization** in data science.



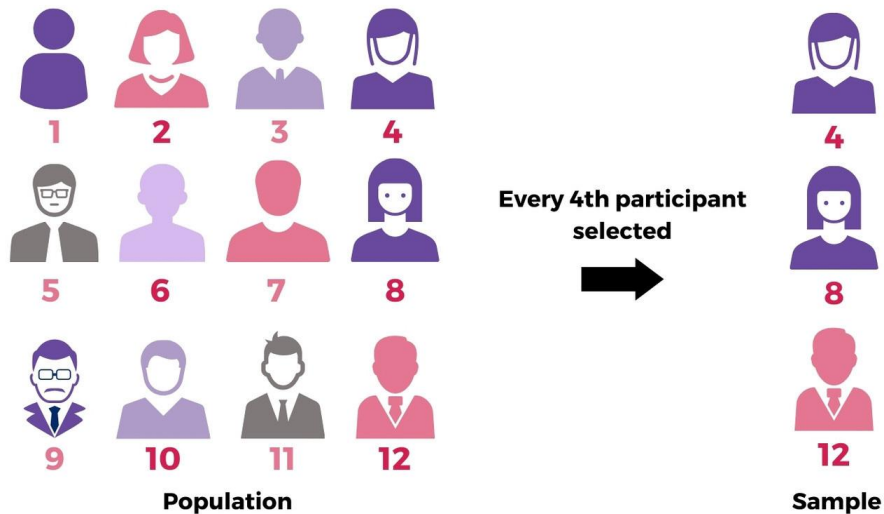
# Sampling Techniques

Different sampling techniques are used to reduce bias:

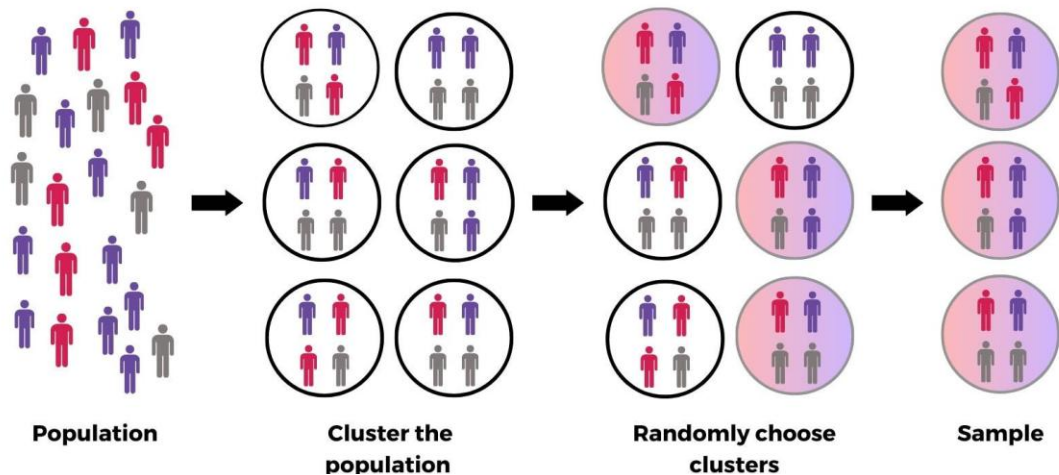
- **Random sampling:** every element has equal chance
- **Stratified sampling:** population divided into groups, sampled proportionally
- **Systematic sampling:** selects elements at **regular intervals** from an ordered population after choosing a random starting point.
- **Cluster sampling:** divides the population into **clusters** and randomly selects **entire clusters** for analysis.

Proper sampling improves **accuracy and reliability of analysis.**

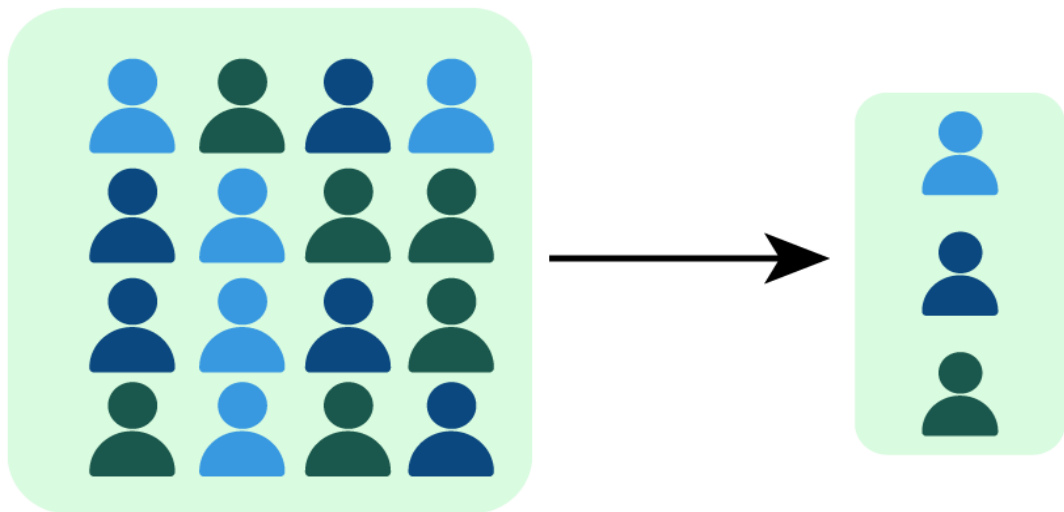
## Systematic Random Sampling



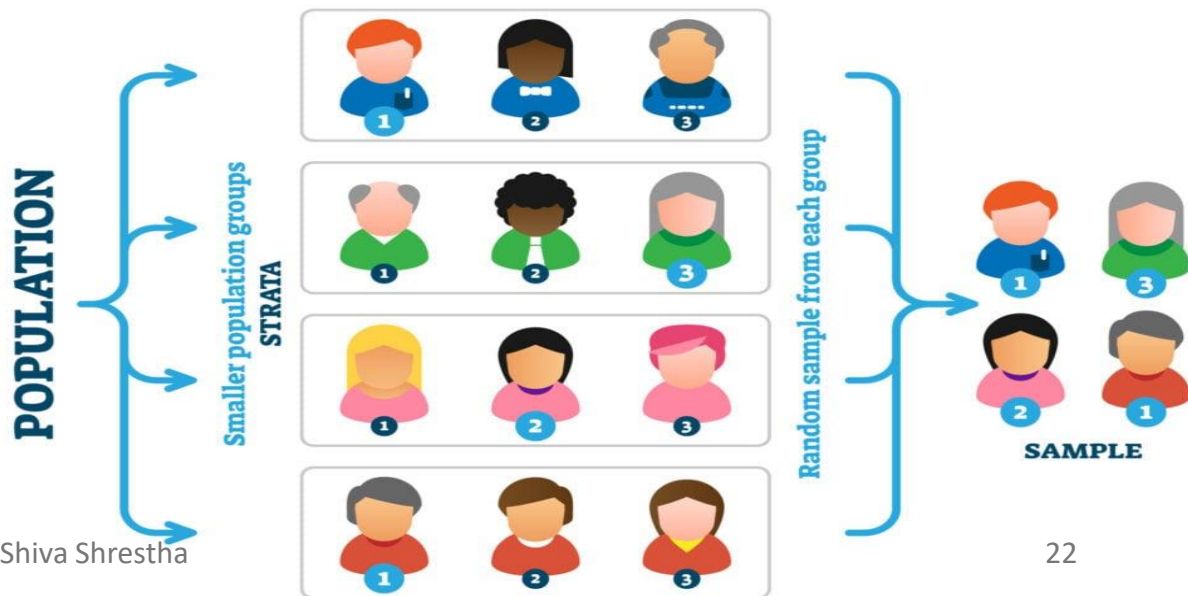
## Cluster Sampling



## Simple Random Sample



## STRATIFIED SAMPLING



# Comparison Summary

| Technique  | How Selection Works    | Main Advantage        | Main Limitation        |
|------------|------------------------|-----------------------|------------------------|
| Random     | Equal chance for all   | Simple, unbiased      | May miss subgroups     |
| Stratified | Random within groups   | Better representation | Needs prior group info |
| Systematic | Every $k$ th item      | Fast, structured      | Periodicity bias       |
| Cluster    | Entire groups selected | Cost-effective        | Less precise           |

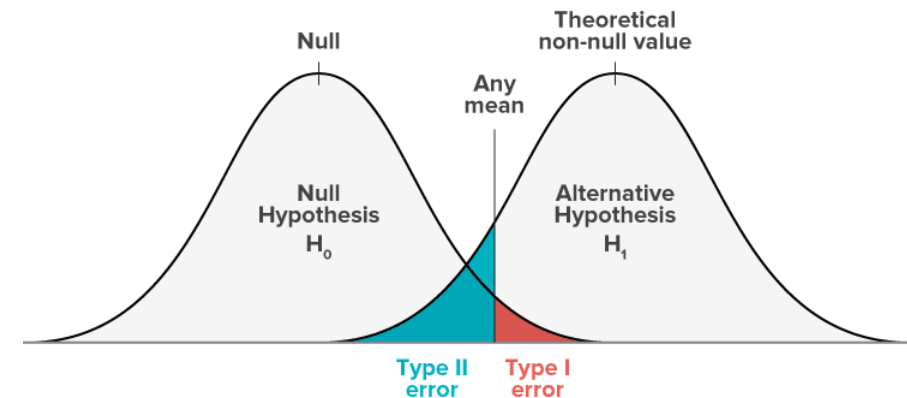
# Hypothesis Testing

**Hypothesis testing** is a statistical method used to make decisions about a population based on sample data.

Key components:

- Null hypothesis ( $H_0$ ): default assumption
- Alternative hypothesis ( $H_1$ ): competing claim

It provides a **systematic decision-making framework**.



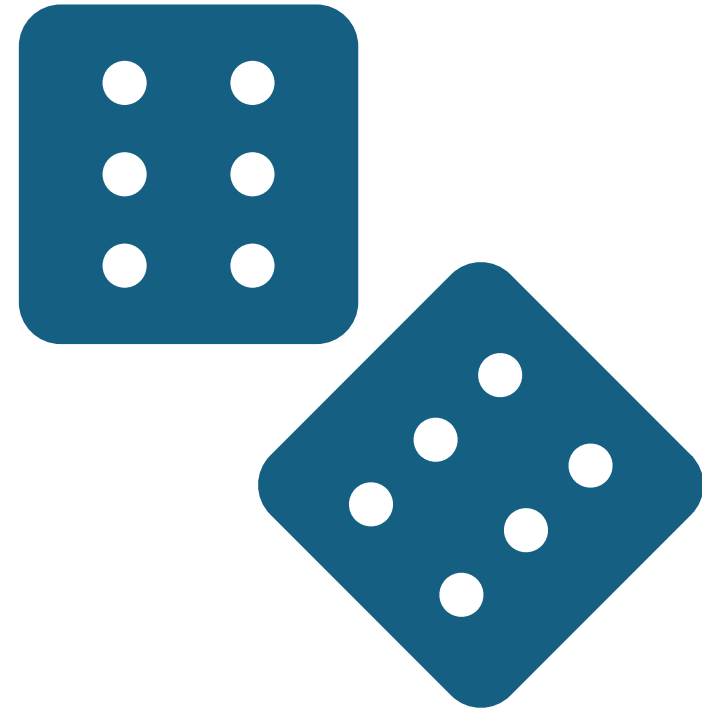


# What is the p-value?

The **p-value** is the probability of obtaining results as **extreme as the observed data**, assuming the **null hypothesis ( $H_0$ )** is true.

- Small p-value  $\rightarrow$  observed result is unlikely under  $H_0$
- Large p-value  $\rightarrow$  observed result is plausible under  $H_0$

**Key idea:** The p-value measures **evidence against the null hypothesis**, not the probability that  $H_0$  is true.



# What is the Significance Level ( $\alpha$ )?

The **significance level ( $\alpha$ )** is a **predefined threshold** used to make decisions in hypothesis testing.

- Common value:  **$\alpha = 0.05$**
- Represents the **maximum acceptable probability of a Type I error** (false positive)

# p-value and Significance

- **Significance level ( $\alpha$ ):** threshold for decision (commonly 0.05)
- **p-value:** probability of observing results assuming  $H_0$  is true

## Decision rule (Very Important):

- If  $p\text{-value} \leq \alpha \rightarrow \text{reject } H_0$
- If  $p\text{-value} > \alpha \rightarrow \text{fail to reject } H_0$

This determines whether results are **statistically significant**.

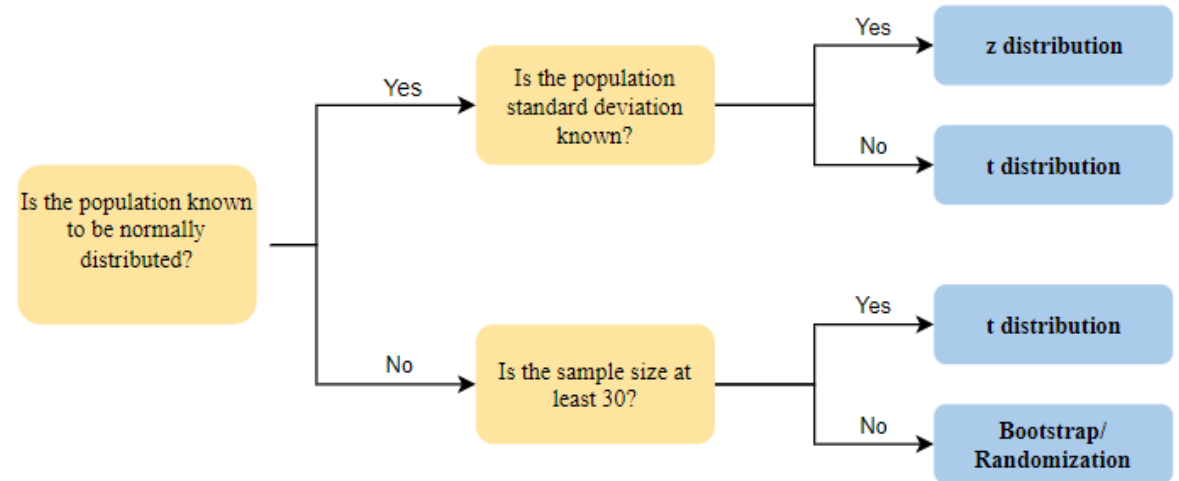
# Overall Statistical Inference

Statistical inference combines:

- Probability theory
- Sampling techniques
- Hypothesis testing

to draw **data-driven conclusions under uncertainty**.

This process is central to **applied statistics and data science**.



## What is Regression Analysis?

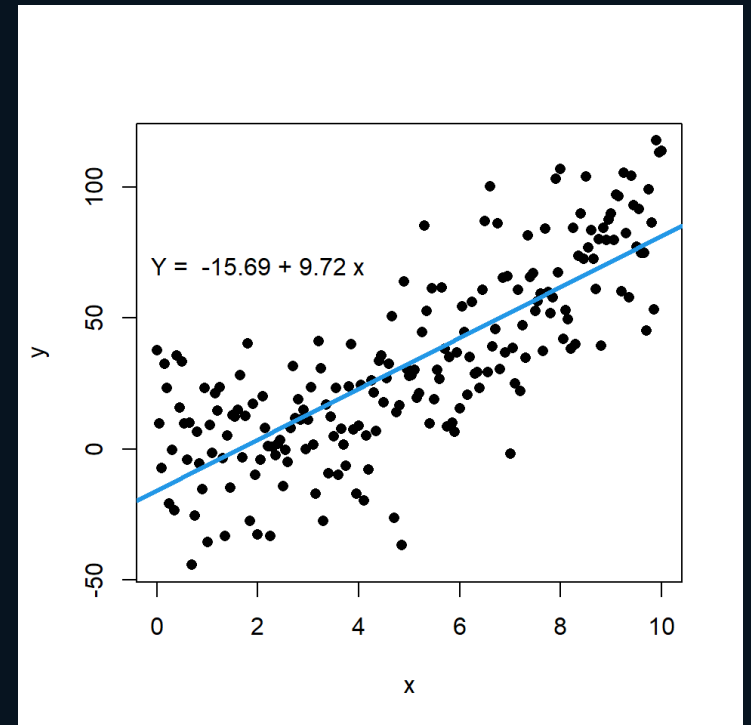
Regression analysis is a statistical technique used to model the relationship between a dependent variable and one or more independent variables.

### Key purposes:

- Understand relationships between variables
- Predict future values
- Explain trends in data

“Regression helps us explain and predict one variable using another.”

## 4.3 Regression and trend analysis using stats models

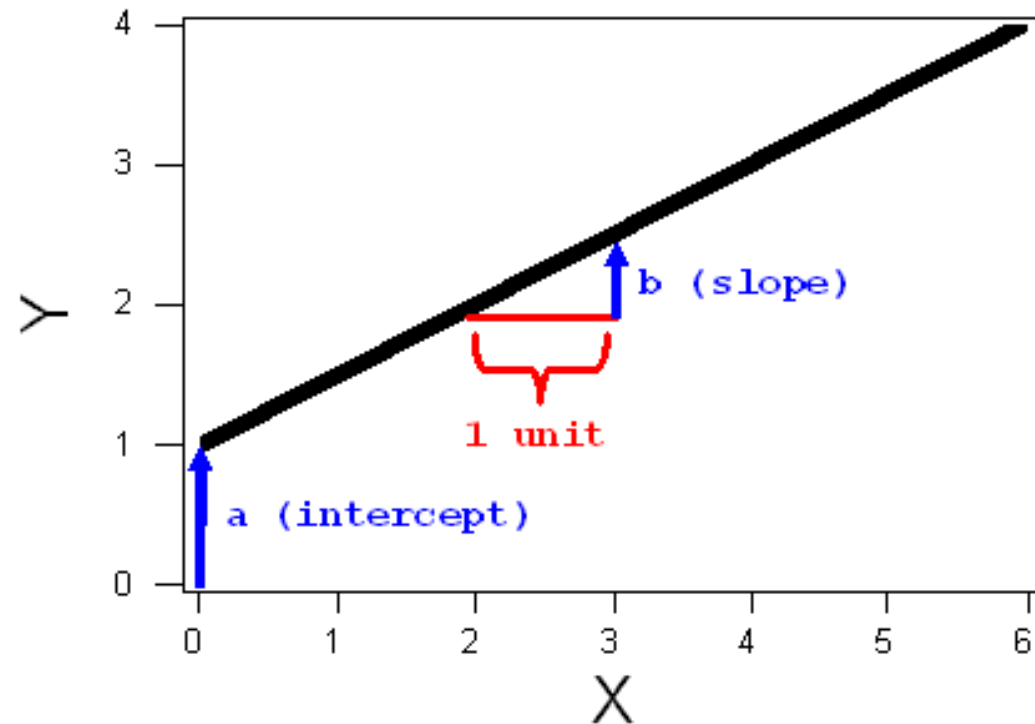


# Simple Linear Regression

- Involves **one independent variable** and **one dependent variable**
- Fits a **best-fit straight line** through data points
- The line represents the average relationship

## Key concepts:

- Slope  $\rightarrow$  rate of change
- Intercept  $\rightarrow$  starting value





**Trend analysis** studies how data values **change over time** to identify long-term movement.

**Key ideas:**

- Increasing trend
- Decreasing trend
- Stable trend

**Used commonly in:**

- Sales analysis
- Economic data
- Performance monitoring

Note: "Trend analysis focuses on direction, not short-term fluctuations."

# Trend Analysis

# Regression vs Trend Analysis

| Aspect       | Regression Analysis                 | Trend Analysis                |
|--------------|-------------------------------------|-------------------------------|
| Focus        | Relationship between variables      | Change over time              |
| Variables    | Dependent and independent variables | Time and one variable         |
| Purpose      | Explanation and prediction          | Identify direction or pattern |
| Role of Time | Optional                            | Mandatory                     |
| Output       | Predictive model                    | Trend line showing movement   |



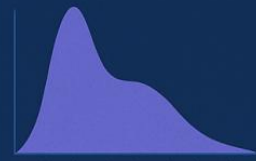
## 4.4 Exploratory data analysis (EDA) using descriptive and inferential methods

### From Raw Data to Early Insights: Preliminary EDA in PD Modelling

Dataset Overview

| Feature           | Missing Values |
|-------------------|----------------|
| Employment Length | 896 (2,75%)    |
| Interest Rate     | 3,116 (9,56%)  |
| Debt to Income    | 9 (0,03%)      |

Probability Distribution



Scatter Plot



Box Plot



Histogram:  
Loan Intent



Crosstab: Home Ownership vs Loan Sta

| Home Ownership | 0   | %   | Rent |
|----------------|-----|-----|------|
| Mortgage       | 41% | 24% | 41%  |
| Own            | 9%  | 3%  | 3%   |
| Rent           | 50% | 18% | 73%  |
|                | 41% | 24% | 50%  |

#### What is Exploratory Data Analysis (EDA)?

**Exploratory Data Analysis (EDA)** is the process of **examining and summarizing data to understand its main characteristics** before formal modeling.

#### EDA focuses on:

- Understanding data structure
- Detecting patterns and anomalies
- Forming hypotheses

# Descriptive vs Inferential EDA

## Descriptive EDA

- Summarizes data
- Uses measures like mean, median, plots
- Spread

*“Descriptive EDA answers: what does the data look like?”*

## Inferential EDA

- Draws conclusions about populations
- Uses probability and hypothesis testing

*“Inferential EDA asks: what can we conclude beyond the sample?”*

Both are essential for complete analysis.

# Descriptive EDA vs Inferential EDA

| Aspect             | Descriptive EDA                    | Inferential EDA                        |
|--------------------|------------------------------------|----------------------------------------|
| Objective          | Describe and summarize data        | Draw conclusions about a population    |
| Question answered  | What does the data look like?      | What can we conclude beyond the data?  |
| Scope              | Limited to the given dataset       | Extends from sample to population      |
| Nature             | Exploratory and observational      | Analytical and decision-oriented       |
| Use of probability | Not required                       | Essential                              |
| Techniques used    | Mean, median, plots, distributions | Hypothesis tests, confidence intervals |
| Data requirement   | Entire dataset or sample           | Sample data                            |
| Output             | Summaries and visual insights      | Statistical decisions and inferences   |
| Assumptions        | Few or none                        | Requires statistical assumptions       |
| Role in workflow   | First step of EDA                  | Later step of EDA                      |

# Common EDA Techniques

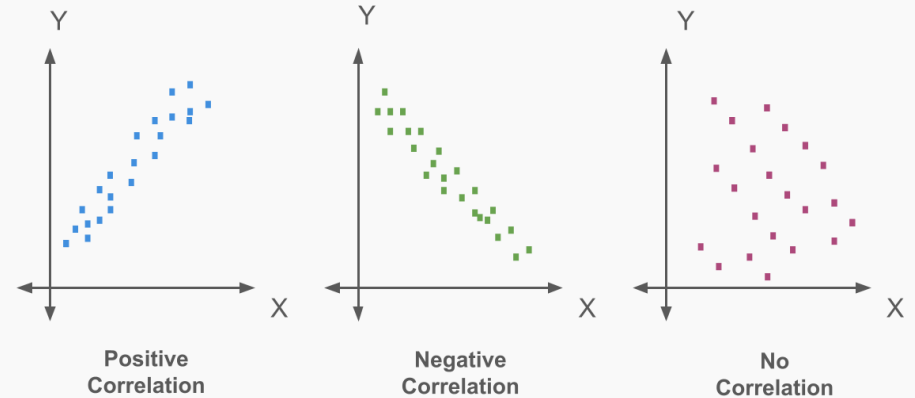
## EDA commonly uses:

- Summary statistics
- Histograms and box plots
- Scatter plots and correlation analysis

## These techniques reveal:

- Distribution shape
- Outliers
- Relationships

### Scatter Plot Correlation Examples

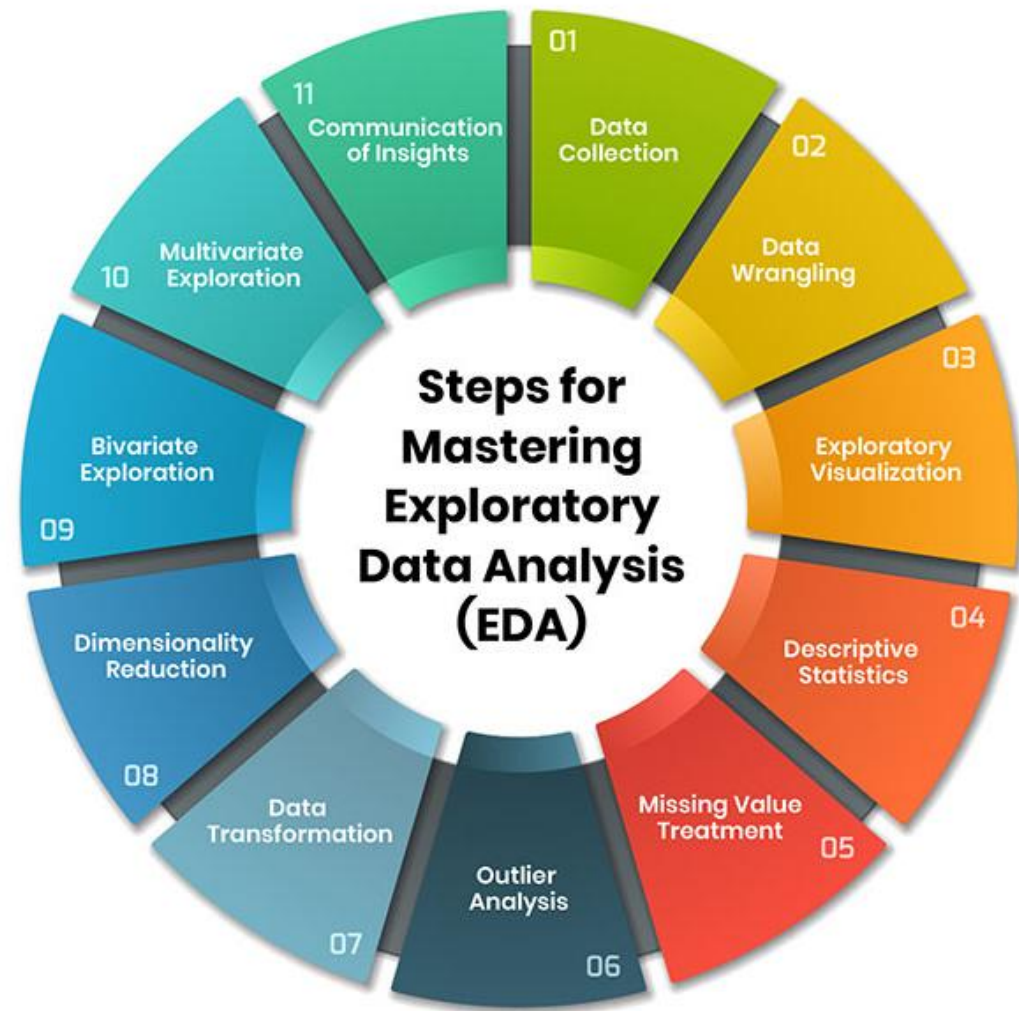


# Why EDA is Important

EDA is important because it:

- Prevents incorrect assumptions
- Identifies data quality issues
- Guides feature selection and modeling

Skipping EDA often leads to **misleading results**.



# EDA in the Data Science Workflow

Data flow:

**Data Collection → Data Cleaning → EDA → Modeling → Decision Making**

**EDA helps to:**

- Understand structure
- Detect anomalies
- Identify relationships
- Guide feature selection

EDA acts as a **bridge** between preparation and modeling.

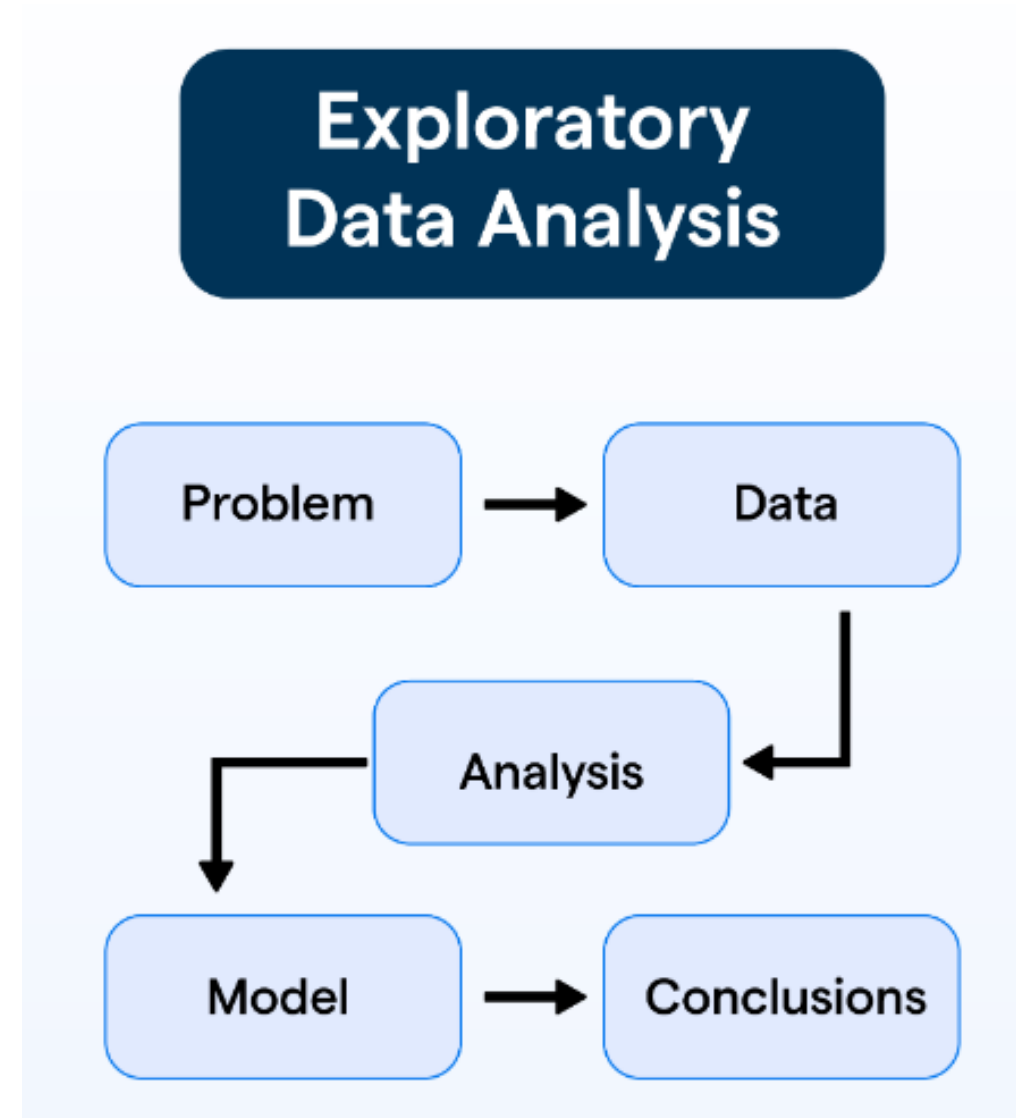
## 4.5 Automation of EDA Workflows Using Python

### What is EDA Automation?

**EDA automation** is the process of using Python programs to **automatically generate exploratory analysis results** without manually repeating steps for every dataset.

It automates tasks such as:

- Summary statistics
- Data distributions
- Missing value analysis
- Correlation analysis
- Basic visualizations



# Why Automate EDA?

## Manual EDA:

- Time-consuming
- Error-prone
- Repetitive

## Automated EDA:

- Faster and consistent
- Reusable across datasets
- Scalable for large or frequent data

Automation is essential in **real-world data science workflows**.

| Automated Data Entry                                                                                                  | VS                    | Manual Data Entry                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------|
|  HIGHLY ACCURATE                   | ACCURACY              | PRONE TO ERRORS            |
|  VERY FAST                         | SPEED                 | TAKES HOURS                |
|  CHEAP MONTHLY COST                | COST                  | EXPENSIVE WAGES            |
|  TACKLES HUGE DATA SETS            | SCALABILITY           | DIFFICULT TO TRAIN PEOPLE  |
|  REQUIRES REPROGRAMMING            | FLEXIBILITY           | REQUIRES TRAINING          |
|  NO LAWS TACKLED                   | REGULATIONS           | DEALS WITH LABOR LAWS      |
|  MATCHES WITH ACCOUNTING SOFTWARE | INTEGRATION           | REQUIRES EXTRA PERSONNEL  |
|  NO ISSUES WITH NEW MANAGEMENT   | ORGANIZATIONAL CHANGE | REQUIRES ORIENTATIONS    |

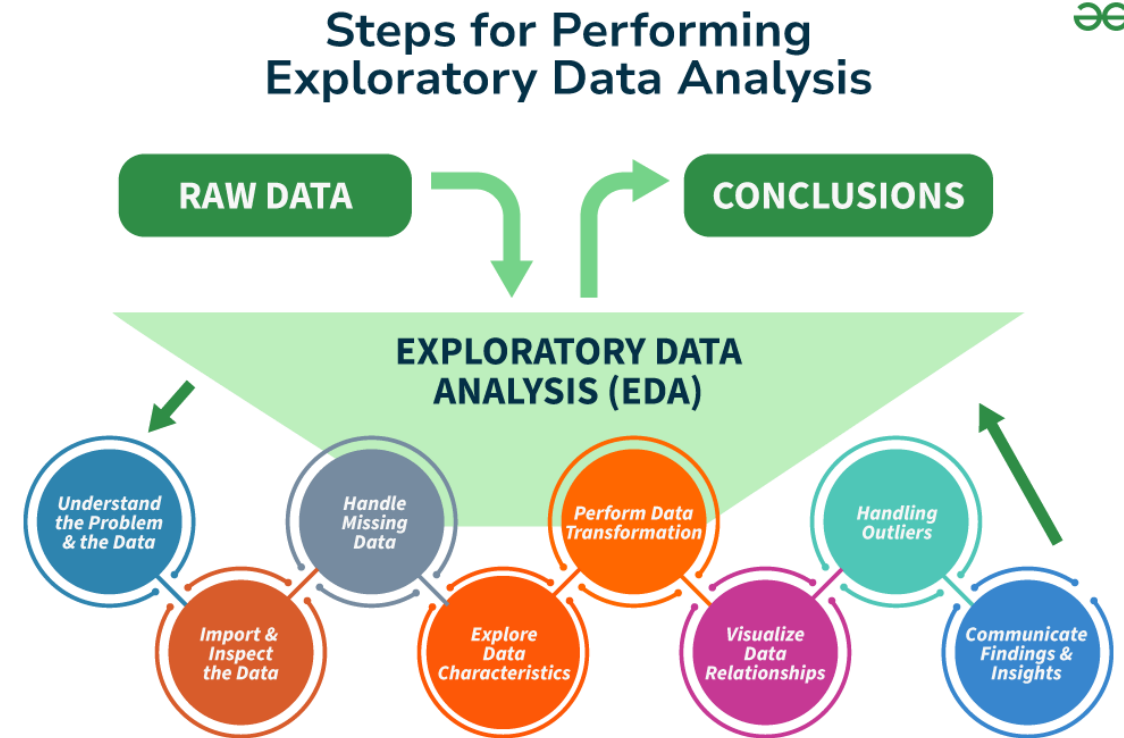


# Typical EDA Steps Suitable for Automation

EDA steps commonly automated include:

- Dataset shape and structure
- Missing value detection
- Summary statistics
- Distribution plots
- Correlation analysis

These steps follow a **fixed and repeatable pattern**.



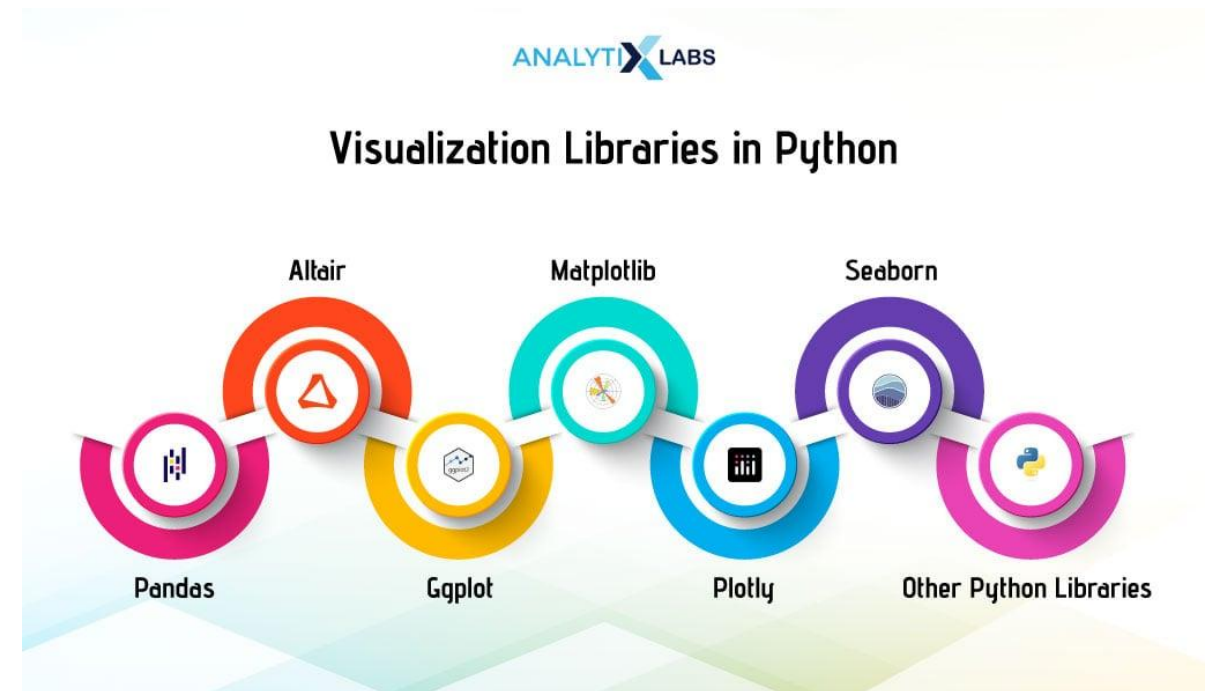
# How Python Enables EDA Automation

Python supports EDA automation through:

- Reusable functions
- Scripts and notebooks
- Data analysis libraries like pandas, numpy, matplotlib

Automation relies on:

- Writing generic analysis functions
- Applying them to multiple datasets

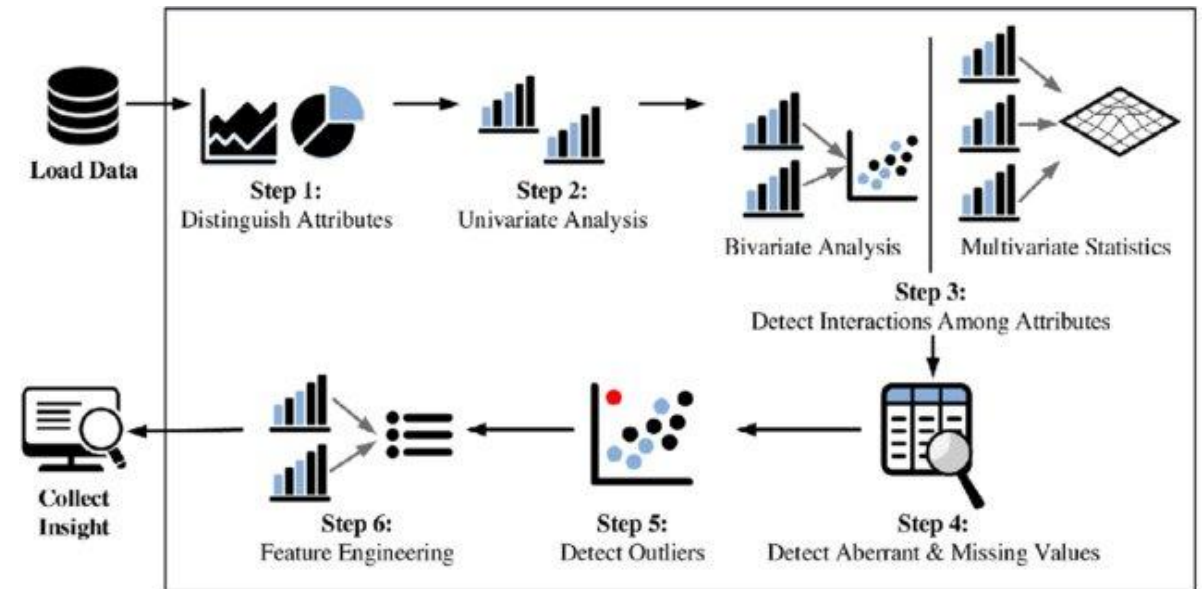


# EDA Automation Workflow

A typical automated EDA workflow:

**Load Data → Analyze Structure →  
Generate Statistics → Create Visuals →  
Save Results**

This workflow can be reused with minimal changes.

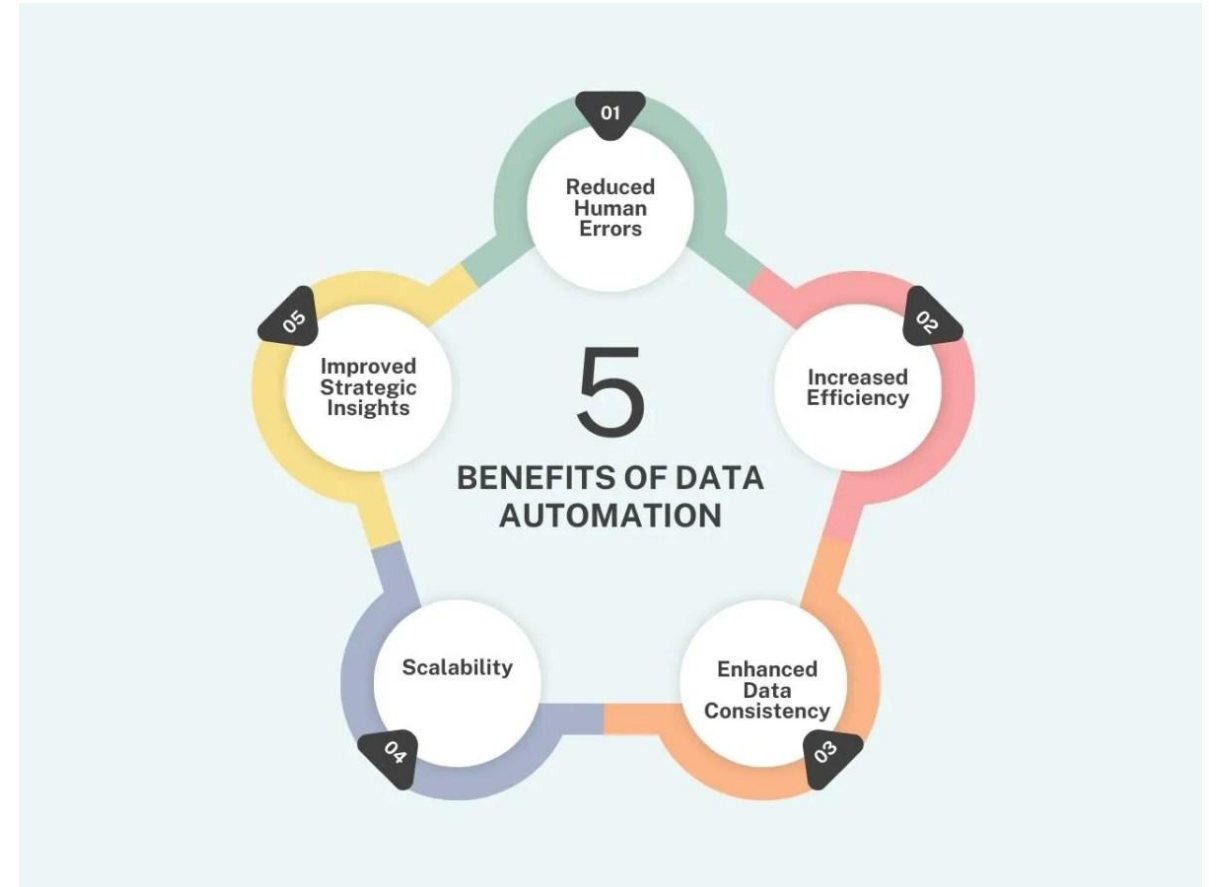


# Benefits of Automated EDA

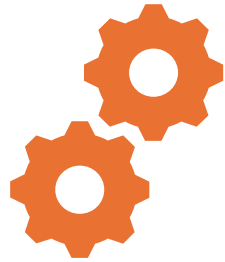
## Automated EDA:

- Saves time
- Reduces human error
- Improves consistency
- Enables rapid dataset comparison

It allows analysts to focus on **interpretation**, not repetition.



# Automated EDA in the Data Science Pipeline



## In practice:

Automated EDA is triggered after data cleaning  
Results guide modeling and decision-making  
Automation supports continuous data updates



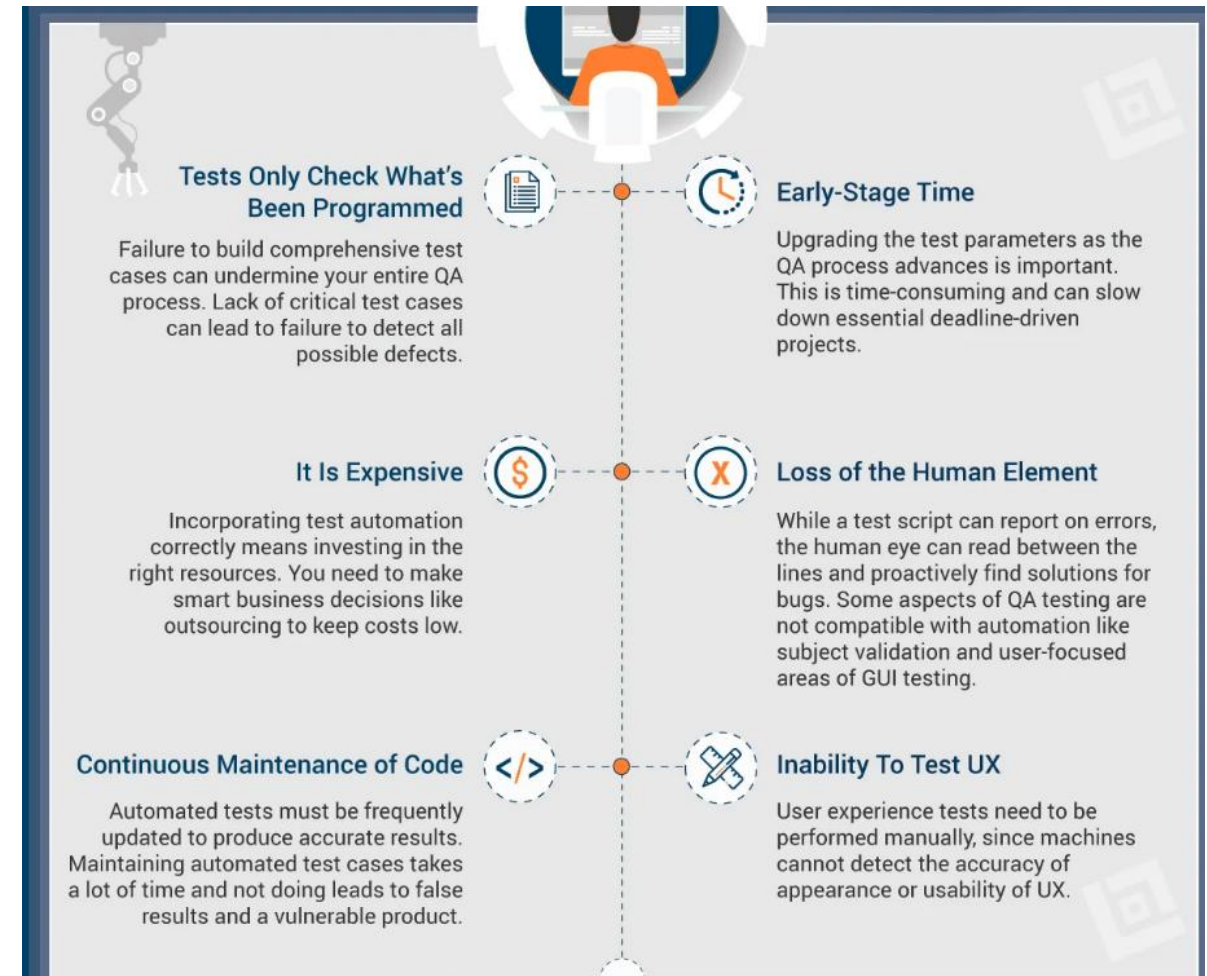
**EDA automation connects data preparation and modeling.**

# Limitations of Automated EDA

## Automation:

- Cannot replace human judgment
- May miss domain-specific insights
- Should support, not replace, manual analysis

Note: "Automation accelerates EDA, but interpretation remains human."



# When to Use Automated EDA

## Use automated EDA when:

- Working with many datasets
- Data updates frequently
- Performing initial data assessment

Avoid full automation for **final analytical decisions**.

## Benefits of Automating Data Intake and Analysis





# Summary of EDA Automation

- EDA automation reduces repetitive analysis work
- Python enables reusable and scalable workflows
- Automation improves speed and consistency
- Human insight is still essential